Lab 7: Fuzz Testing with Hypothesis

SRN: PES1UG24CS248 | Repo: L6_L7_03_coffee_shop_order (Lab 7)

Step 2: Running the buggy processor.py

Entering the valid code `mocha` crashes with `KeyError: 'MOC'`, because the menu key is misspelt as `"MOc"`.

Step 2 — buggy processor.py

```
$ python processor.py
==== Running lookup_drink_code ====
Enter drink code prefix (e.g. LAT, CAP, ESP): mocha
Traceback (most recent call last):
  File "processor.py", line 41, in <module>
    main()
    ~~~~^^
  File "processor.py", line 27, in main
    price = lookup_drink_code(drink_input)
  File "processor.py", line 10, in lookup_drink_code
    return menu_prices[code_prefix]
           ~~~~~~~~~~~^^^^^^^^^^^^^
KeyError: 'MOC'
```

Step 4: Fuzz tests failing on the buggy code

All 6 Hypothesis tests in `test_processor.py` fail. The second screenshot lists the minimal failing inputs Hypothesis found.

Step 4 — fuzz tests failing (buggy code)

```
$ python -m pytest test_processor.py --tb=no -v
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0 -- /usr/local/bin/python
hypothesis profile 'default'
rootdir: ~/L6_L7_03_coffee_shop_order/Lab 7
plugins: platformdirs-4.13.5, anyio-4.15.1, cov-7.1.0, hypothesis-6.170.15
collecting ... collected 6 items

test_processor.py::test_lookup_drink_code_no_crash FAILED                [ 16%]
test_processor.py::test_lookup_drink_code_valid_prefix_any_case FAILED   [ 33%]
test_processor.py::test_calculate_average_shot_cost_safe FAILED          [ 50%]
test_processor.py::test_calculate_average_shot_cost_numeric_csv FAILED   [ 66%]
test_processor.py::test_parse_order_modifiers_safe FAILED                [ 83%]
test_processor.py::test_parse_order_modifiers_separator_heavy FAILED     [100%]

=========================== short test summary info ============================
FAILED test_processor.py::test_lookup_drink_code_no_crash - ExceptionGroup: H...
FAILED test_processor.py::test_lookup_drink_code_valid_prefix_any_case - Asse...
FAILED test_processor.py::test_calculate_average_shot_cost_safe - ExceptionGr...
FAILED test_processor.py::test_calculate_average_shot_cost_numeric_csv - Exce...
FAILED test_processor.py::test_parse_order_modifiers_safe - ExceptionGroup: H...
FAILED test_processor.py::test_parse_order_modifiers_separator_heavy - Assert...
============================== 6 failed in 0.17s ===============================
```

Step 4 — minimal failing inputs found by Hypothesis

```
$ python -m pytest test_processor.py 2>&1 | grep -E "AssertionError: .*crashed on|isfinite>\(inf\)" | sort -u
+  and   False = <built-in function isfinite>(inf)
AssertionError: calculate_average_shot_cost crashed on '' with: ValueError: could not convert string to float: ''
AssertionError: calculate_average_shot_cost crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
AssertionError: lookup_drink_code crashed on '' with: KeyError: ''
AssertionError: lookup_drink_code crashed on None with: AttributeError: 'NoneType' object has no attribute 'upper'
AssertionError: lookup_drink_code crashed on valid code 'MoC' with: KeyError: 'MOC'
AssertionError: parse_order_modifiers crashed on '' with: ValueError: not enough values to unpack (expected 2, got 1)
AssertionError: parse_order_modifiers crashed on None with: AttributeError: 'NoneType' object has no attribute 'split'
```

Step 5: Running the fixed processor.py

The same `mocha` input now returns $5.00. Invalid CSV entries and malformed modifiers are skipped instead of crashing.

Step 5 — fixed processor.py

```
$ python processor.py
==== Running lookup_drink_code ====
Enter drink code prefix (e.g. LAT, CAP, ESP): mocha
Base Price: $ 5.0

==== Running calculate_average_shot_cost ====
Enter CSV of shot prices (e.g. 1.0, 1.5, 2.0): 1.0, abc, , 2.0
Average Shot Price: $ 1.5

==== Running parse_order_modifiers ====
Enter modifiers (e.g. milk:oat;syrup:vanilla): milk:oat;bad;syrup:vanilla
Parsed Modifiers: {'milk': 'oat', 'syrup': 'vanilla'}
```

Step 6: Fuzz tests passing on the fixed code

All 6 Hypothesis tests pass against the hardened `processor.py`.

Step 6 — fuzz tests passing (fixed code)

```
$ python -m pytest test_processor.py -v
============================= test session starts ==============================
platform linux -- Python 3.13.18, pytest-9.1.1, pluggy-1.6.0 -- /usr/local/bin/python
hypothesis profile 'default'
rootdir: ~/L6_L7_03_coffee_shop_order/Lab 7
plugins: platformdirs-4.13.5, anyio-4.15.1, cov-7.1.0, hypothesis-6.170.15
collecting ... collected 6 items

test_processor.py::test_lookup_drink_code_no_crash PASSED                [ 16%]
test_processor.py::test_lookup_drink_code_valid_prefix_any_case PASSED   [ 33%]
test_processor.py::test_calculate_average_shot_cost_safe PASSED          [ 50%]
test_processor.py::test_calculate_average_shot_cost_numeric_csv PASSED   [ 66%]
test_processor.py::test_parse_order_modifiers_safe PASSED                [ 83%]
test_processor.py::test_parse_order_modifiers_separator_heavy PASSED     [100%]

============================== 6 passed in 0.40s ===============================
```

Bugs found by fuzzing and fixes applied

Function	Input	Bug	Fix
`lookup_drink_code`	None input	AttributeError on .upper()	Return None for non-string input
`lookup_drink_code`	Unknown / empty code	KeyError	Use dict.get(), return None
`lookup_drink_code`	'MOC…' (mocha)	KeyError: menu key typo "MOc"	Corrected key to "MOC"
`calculate_average_shot_cost`	None input	AttributeError on .split()	Return 0.0 for non-string input
`calculate_average_shot_cost`	'' / '1,,2' / 'abc'	ValueError in float()	Skip empty and non-numeric entries
`calculate_average_shot_cost`	No valid prices	ZeroDivisionError	Return 0.0 when list is empty
`calculate_average_shot_cost`	'inf', 'nan', huge values	Non-finite average	Skip non-finite / negative values; divide before summing
`parse_order_modifiers`	None input	AttributeError on .split()	Return {} for non-string input
`parse_order_modifiers`	'' / 'bad' (no colon)	ValueError: not enough values to unpack	Skip items without ':'
`parse_order_modifiers`	'a:b:c' (extra colon)	ValueError: too many values to unpack	split(':', 1)
`parse_order_modifiers`	':oat' (empty key)	Empty-string key stored	Skip items with empty key